

Calibration - Theory

Coordinate Frames

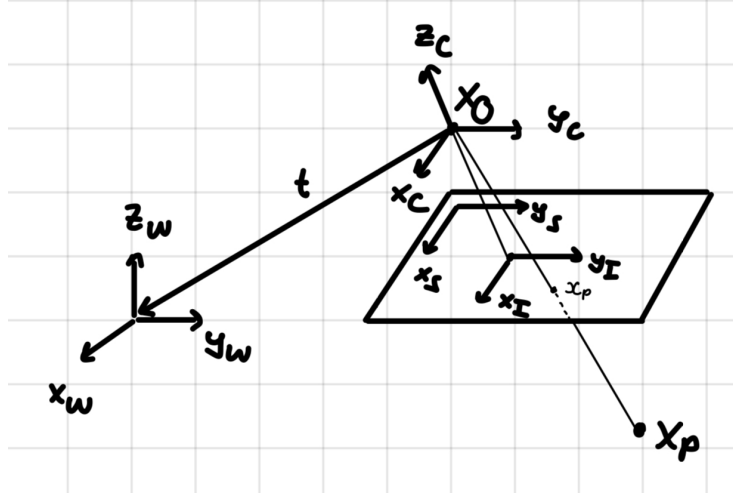


Figure 1: Coordinate frames

world : w

$$X_P = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}, \tilde{X}_P = \tilde{X}_P^w = \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}, X_O = \begin{bmatrix} o_x \\ o_y \\ o_z \end{bmatrix}$$

camera : c

$$X_P^c = \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}, \tilde{X}_P^c = \begin{bmatrix} \tilde{X}_c \\ \tilde{Y}_c \\ \tilde{Z}_c \\ \tilde{W}_c \end{bmatrix}, \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \begin{bmatrix} \tilde{X}_c/\tilde{W}_c \\ \tilde{Y}_c/\tilde{W}_c \\ \tilde{Z}_c/\tilde{W}_c \end{bmatrix}, t = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}, t = -R * X_O$$

image plane : I

sensor : s

$$x_p = \begin{bmatrix} u \\ v \end{bmatrix}, \tilde{x}_p = \tilde{x}_p^s = \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix}, \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \tilde{u}/\tilde{w} \\ \tilde{v}/\tilde{w} \end{bmatrix}$$

Projection Geometry

Projection from world to camera

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = [R \quad -R * X_O] * \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$X_P^c = R * (X_P - X_O)$$

$$X_P = R^T * X_P^c + X_O$$

Projection from camera to sensor

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$$

$$\tilde{x}_p = K * X_P^c$$

$$X_P^c = \lambda * K^{-1} * \begin{bmatrix} x_p \\ 1 \end{bmatrix}$$

Note: All points along a ray that passes through the camera origin map to the same pixel coordinates. Going from pixel coordinates to 3d coordinates has an infinite solution space with 1 DoF, the distance along the ray.

Projection from world to sensor

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} * [R \quad -R * X_O] * \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\tilde{x}_p = K * R * (X_p - X_O)$$

$$X_p = \lambda * (K * R)^{-1} * \tilde{x}_p + X_O$$

when $Z = 0$, the relation simplifies to

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} r1 & r2 & -R * X_O \end{bmatrix} * \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

$$\tilde{x}_p = H * \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} \tilde{X} \\ \tilde{Y} \\ \tilde{W} \end{bmatrix} = H^{-1} * \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

Note: Restricting the object points to lie on a plane, the mapping from world to pixel coordinates becomes invertible, since it maps from 2d to 2d and doesn't have loss of information.

Distortion

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} X_c/Z_c \\ Y_c/Z_c \end{bmatrix}$$

$$\begin{bmatrix} x_d \\ y_d \end{bmatrix} = \begin{bmatrix} x + \Delta x(x_d, y_d, q) \\ v + \Delta y(x_d, y_d, q) \end{bmatrix}$$

$$\begin{bmatrix} u_d \\ v_d \\ 1 \end{bmatrix} = K * \begin{bmatrix} x_d \\ y_d \\ 1 \end{bmatrix}$$

OpenCV models radial and tangential distortion with five parameters:

$$q = (k1, k2, p1, p2, k3)$$

$$\Delta x_{radial}(x, y, q) = x * (k1 * r^2 + k2 * r^4 + k3 * r^6)$$

$$\Delta y_{radial}(x, y, q) = y * (k1 * r^2 + k2 * r^4 + k3 * r^6)$$

$$\Delta x_{tangential}(x, y, q) = 2 * p1 * x * y + p2 * (r^2 + 2 * x^2)$$

$$\Delta y_{tangential}(x, y, q) = p1 * (r^2 + 2 * y^2) + 2 * p2 * x * y$$